5_2 DFT 1 :

Given $y = [0, 1, 0, -1]$

$N = 4$

$q = \frac{N}{2} = 2 \quad [N = \text{even}]$

We should calculate

a_0, a_1, a_2, b_1, b_2

$b_2 = 0$ ∵ since N is even

$$a_0 = \frac{1}{N} \sum_{t=0}^{N-1} y_t$$

$$= \frac{1}{4} (0 + 1 + 0 - 1) = 0$$

$$a_1 = \frac{2}{N} \sum_{t=0}^{N-1} y_i \cos\left(\frac{2\pi t}{4}\right)$$

$$= \frac{1}{2} \sum_{t=0}^{3} y_{0,i} \cos \frac{\pi t}{2}$$

$$= \frac{1}{2} [0.(1) + 1 \cdot 0 + 0 \cdot (-1) + (-1) \times 0]$$

$$= 0$$

$$a_2 = \frac{1}{4} \sum_{t=0}^{3} y_t (-1)^t$$

$$a_2 = \frac{1}{4} [\ 0 - 1 + 0 + 1] = 0$$

$$b_1 = {}^2/_4 \sum_{t=0}^{3} y_t \sin\left(\frac{2\pi t}{4}\right)$$

$$b_1 = \frac{1}{2} [0(0) + 1(1) + 0(0) + (-1)(-1)]$$

$$b_1 = 1$$

& $b_2 = 0$